OFFICIAL WORKFLOW DOCUMENTATION

Workflow: End-to-End Operational Lifecycle

Current Production State Architecture Across Student Portal, Faculty Admin & Cloud AI

Project Title:	CareerForge AI (Current Production State)
Portals Documented:	1. Student Autonomous Learning Portal (index.html) 2. Faculty Administration & Placement Governance Portal (admin.html)
Active AI Model:	Google Gemini 3.6 Flash (models/gemini-3.6-flash)
Backend Database:	Supabase PostgreSQL (auth.users + public.student_states)
Visual Assets:	25 Curated Local 16:9 Workstation PNG Images (0ms instant loading)
Document Type:	Full System Lifecycle & Data Flow Specification (Exam & Viva Reference)

Current Workflow State: This document details the live, fully integrated system state. It reflects the upgrade of Technical and HR interviews to Google Gemini 3.6 Flash, the removal of third-party image generation APIs in favor of 25 local 16:9 workstation assets, real-time Supabase PostgreSQL cloud sync, strict anti-gibberish scoring, and verified placement endorsement governance.

1. System Architecture & Dual-Portal Pipeline

CareerForge operates on a modern, decoupled client-server architecture consisting of three core pillars:

- **1. Student Reactive Portal (index.html):** A single-page application (SPA) powered by a hash-based reactive router (js/router.js). Handles self-discovery, resume building, interview practice, and certificate unlocks without page reloads.
- **2. Faculty Admin Portal (admin.html):** An administrative management console (js/admin.js, css/admin.css) providing cohort analytics, student dossiers, certificate approvals, and CSV exports.
- **3. Cloud & Intelligence Services:** Supabase PostgreSQL (managing authentication and real-time state sync) and Google Gemini 3.6 Flash (generating dynamic interview questions and grading mock sessions).

Pillar	Component	Technology Used	Operational Function
Frontend SPA	Student Portal (index.html)	HTML5, CSS3, Vanilla JS (ES6+)	Reactive UI rendering, 3D Canvas flight tunnel, client state machine (S).
Faculty Portal	Admin Portal (admin.html)	Admin CSS, Admin JS, Supabase Client	Direct querying of live students, search/filtering, dossier inspector, endorsement actions.
Cloud AI	Gemini Engine (js/ai.js)	Google Gemini 3.6 Flash REST API	Generates 5 domain-tailored technical MCQs, 5 HR behavioral MCQs, and evaluates written mock answers.
Cloud Database	Supabase Backend (supabase.js)	PostgreSQL 15 (Auth + JSONB)	Authenticates users via JWT, stores state snapshots in student_states, syncs portals.

2. Student Portal: Complete Step-by-Step Workflow

Stage	Route & Screen	User Journey Action	System Mechanism & State Mutation
Stage 1	#/splash #/signup #/login	Student registers account or signs in.	Calls supabase.auth.signUp() or signInWithPassword(). Returns JWT session token. Initializes state record in student_states table.
Stage 2	#/profile-setup	Inputs Full Name, College Roll Number, Dept, Year.	Saves profile data into S.profile and upserts to Supabase so the Faculty Admin portal can display their collegiate identity.
Stage 3	#/domains #/assessment	Explores 25 domains; completes 12 diagnostic questions.	Evaluates problem solving and cognitive tendencies across a multi-attribute matrix scoring affinity for all 25 career tracks.
Stage 4	#/result #/future	Reviews Top 5 matches with glowing borders; selects domain.	Locks S.domain. Clicking 'View Future' renders the dedicated 16:9 workstation PNG image from img/ instantly (0ms delay).
Stage 5	#/dashboard	Accesses central operational headquarters.	Calculates and renders circular animated **Readiness Index (0-100)**, streak tracker, accumulated XP points, and practice module launchpads.
Stage 6	#/resume #/resume-builder	Scans resume draft or builds one with 3D Warp Canvas.	ATS scanner generates score & missing domain keywords. 3D flight canvas projects 110 vector stars with type-reactive velocity surges (1.0 to 10.0).
Stage 7	#/interview-home (Technical & HR)	Takes AI-powered Technical and HR interview drills.	Calls Gemini 3.6 Flash via REST. Synthesizes 5 domain-tailored technical MCQs or 5 HR behavioral MCQs with in-depth recruiter explanations.
Stage 8	#/mock-home (AI Mock Interview)	Types free-form paragraph answers to interview prompts.	Gemini evaluates response. Anti-gibberish filter assigns 0% to keyboard mash. Genuine answers receive 4-tier score (Relevance, Communication, Technical, Structure).

Stage 9	#/reflection #/rewards	Logs qualitative reflection; views earned badges.	Awards completion XP points, updates streak, unlocks milestone badges in Rewards Vault.
Stage 10	#/certificate #/cert-preview	Applies for Placement Readiness Endorsement.	When Readiness Index is >= 75%, certificate application unlocks. Sets status to 'Pending' in Supabase for faculty review.

3. Faculty Admin Portal: Complete Workflow

The Faculty Administration Portal (admin.html) gives placement officers and faculty real-time governance:

Phase 1: Real-Time Institutional Intelligence Dashboard

- **Aggregate KPI Metrics:** Computes Total Registered Students, Institutional Average Readiness Index, Placement-Eligible Percentage (students with readiness >= 75%), and Pending Endorsements counter in real-time from Supabase.
- **Domain Enrollment Matrix:** Visual breakdown displaying how students are distributed across Computer Science, Data Science, Cloud, Cybersecurity, and 21 other career tracks.

Phase 2: Live Student Cohort Table (Zero Dummy Data)

- **Direct Supabase PostgreSQL Sync:** On page load, admin.js connects to Supabase and queries the student_states table. Zero dummy/seed records are displayed.
- **Multi-Criteria Search & Filtering:** Faculty can filter records instantaneously by:
- Free-text search bar (Student Name or Roll Number)
- Academic Department (CS, IT, ECE, Mechanical)
- Academic Year (1st, 2nd, 3rd, 4th Year)
- Target Career Domain (all 25 domains)
- Certificate Endorsement Status (Pending, Approved, Rejected, Not Eligible)

Phase 3: Interactive Student Dossier Modal

- Clicking any student in the cohort table opens an in-depth **Student Dossier Modal** showing:
- Breakdown of ATS Resume score, attempts, and uploaded date.
- Gemini Technical Interview score vs Gemini HR Interview score.
- Gemini AI Mock Interview 4-tier score and written hiring feedback.
- Student self-reflections and earned XP points.

Phase 4: One-Click Certificate Governance & CSV Accreditation Export

- **Instant Endorsement Engine:** Faculty can click **'Approve'** or **'Reject'** on a pending certificate directly from the dossier modal. The decision updates Supabase immediately, unlocking the verified certificate seal in the student's portal.
- **Accreditation CSV Export:** One-click button exports a consolidated institutional CSV report formatted for University Placement Drives, NAAC/ABET accreditation audits, and department review meetings.

Admin Step	Action	Technical Mechanism	Outcome
1	Open admin.html	Establishes Supabase client; queries student_states.	Renders live institutional KPIs and enrollment distribution.
2	Search & Filter	Executes multi-field filter predicate in memory.	Instantly pinpoints students by department, year, or readiness score.
3	Open Dossier	Reads full state_data JSONB object.	Displays resume attempts, interview breakdowns, and AI feedback.
4	Approve Certificate	Executes update on state_data.cert.status in Supabase.	Student's certificate is instantly unlocked with verified seal.
5	Export CSV	Parses table data into structured CSV format.	Downloads institutional placement report for university accreditation.

4. End-to-End Data & AI Pipeline Specification

A. Google Gemini 3.6 Flash Integration Pipeline (Single-Trip Credit Preservation):

1. **Dynamic Question Generation (Technical & HR):** When a student starts an interview, generateInterviewQuestionsWithAI() sends a single compact prompt to gemini-3.6-flash:generateContent requesting 5 domain-tailored MCQs with options, correct answer index, explanation, and tips in strict JSON format (~400 tokens).
2. **AI Mock Evaluation & Anti-Gibberish Engine:**

- *Client Pre-Validation:* isGibberishText() checks vowel ratio (< 12%) and consecutive consonants. Nonsense (e.g. 'sfdfdsfsnfbsjdfsdjf') gets **0%** instantly without burning API credits.
- *LLM Evaluation:* Genuine answers are graded across 4 categories: Relevance, Communication, Technical Depth, and Structure.

3. **Zero-Crash Fallback:** If internet drops or API quota is reached, a curated domain fallback smoothly activates, ensuring the UI never crashes.

B. Supabase Real-Time State Sync Pipeline:

1. **Client State Mutation:** All student actions update the central reactive state tree S in js/router.js.
2. **Dual-Layer Persistence:** saveState() writes to browser localStorage first (for 0ms local speed), then triggers an asynchronous UPSERT into PostgreSQL public.student_states.
3. **Administrative Reflection:** Faculty opening admin.html read the live state immediately. When faculty approves a certificate, the database update reflects on the student's screen on their next action or refresh.

C. 25 Domain Image Mapping Pipeline (100% Offline Asset Delivery):

- **Local Asset Delivery:** Replaced external image generation APIs with 25 curated, high-definition 16:9 PNG images stored locally in img/.
- **Mapping Engine:** In js/ai.js, DOMAIN_IMAGE_MAP directly maps all 25 domain IDs (uiux, aiml, frontend, cyber, etc.) to their respective image files. Renders in 0ms with smooth CSS opacity transition and safe onerror fallback.

Verification Statement: This workflow document accurately reflects the current, fully implemented production architecture of CareerForge AI. All systems—including Google Gemini 3.6 Flash question generation and mock evaluation, Supabase PostgreSQL authentication and state synchronization, 25 local domain workstation assets, and the Faculty Admin command center—are fully operational and synchronized.